Fire Incident Trend Analysis

Singapore Fire Statistics & SafeEdge Detection Context

Prepared for: Firefighters & Fire Safety Officers

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SafeEdge

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

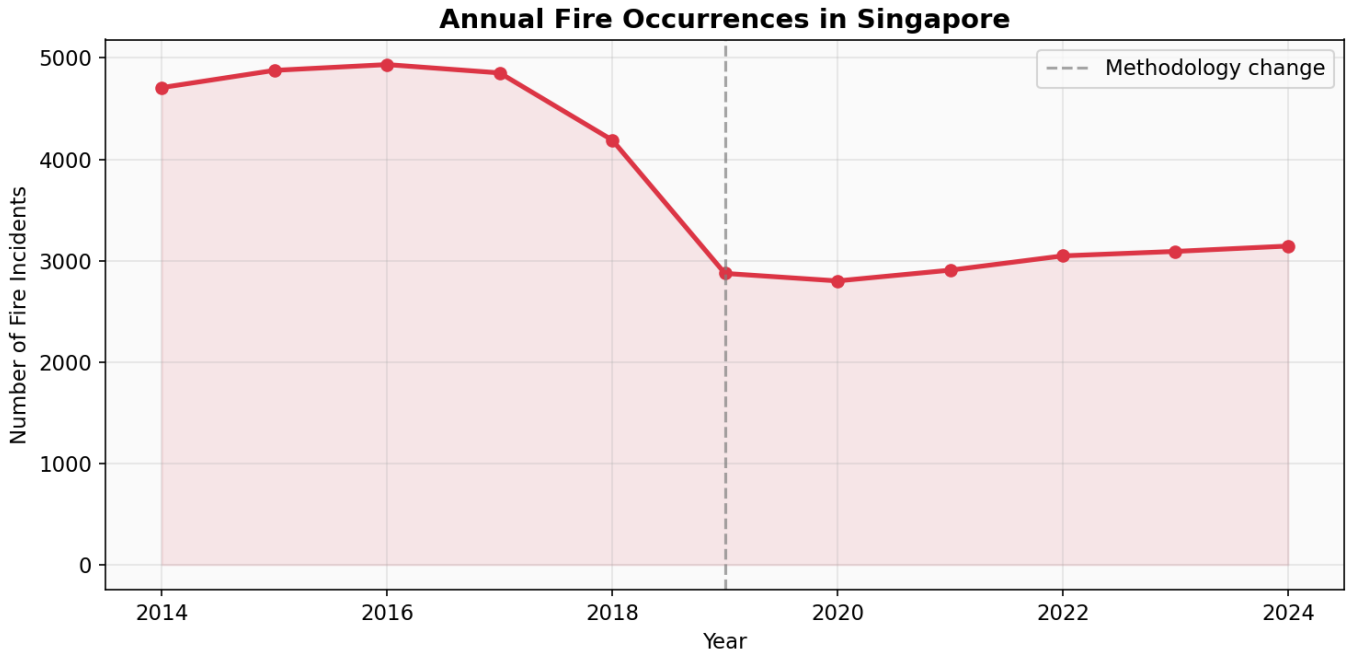
Executive Summary

In 2024, Singapore experienced a total of 3,144 fire incidents, reflecting a 1.7% increase from the previous year. Residential fires accounted for a significant portion of these incidents, totaling approximately 1,160, underscoring the ongoing concern for fire safety in domestic settings. The year also saw 53 reported fire-related injuries and a single fatality, indicating a need for continued vigilance and preventive measures to enhance public safety and reduce the risk of fire-related harm.

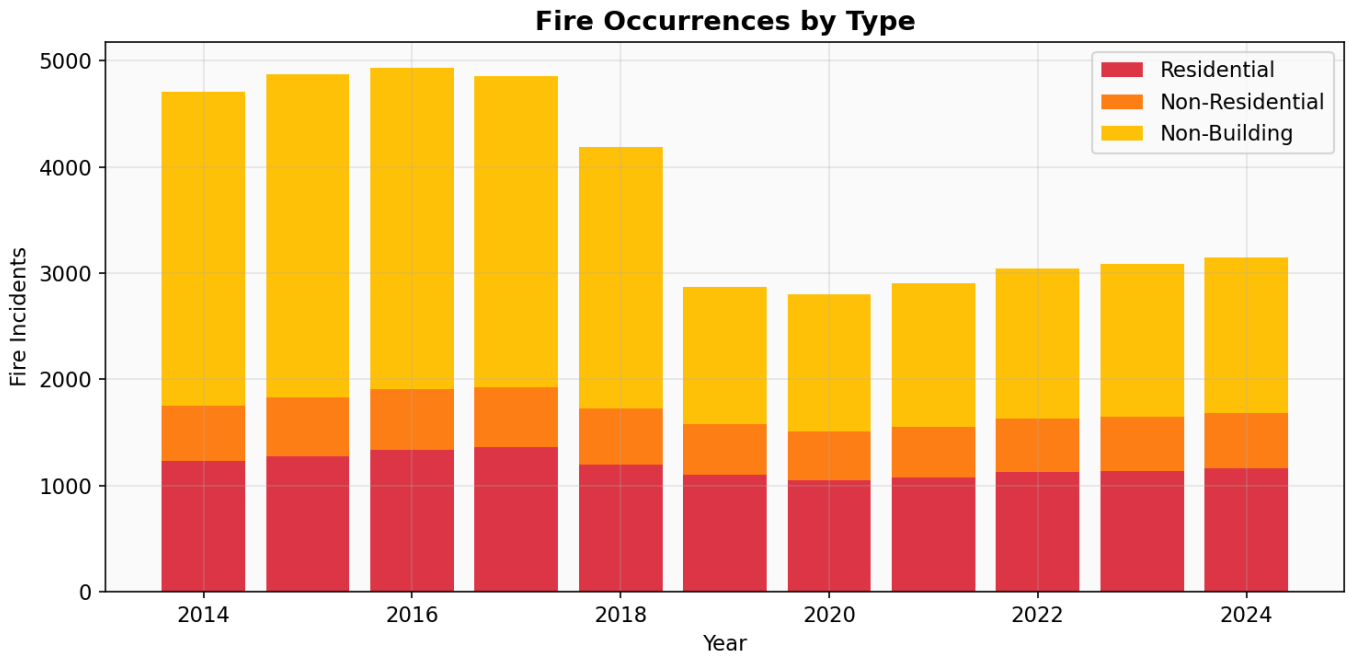
3,144	2024	+1.7%	1
Total Fires	Year	YoY Change	Fatalities

Annual Fire Occurrence Trends

Fire incidents have exhibited significant fluctuations from 2014 to 2024, with reported occurrences ranging from 2,800 to 4,934. A notable decline in incidents was observed around 2019, attributed to methodological changes in data collection and reporting practices. However, following this period, there has been a gradual increase in fire incidents post-2020, indicating a potential resurgence in fire activity that warrants further investigation and proactive measures by fire safety professionals.



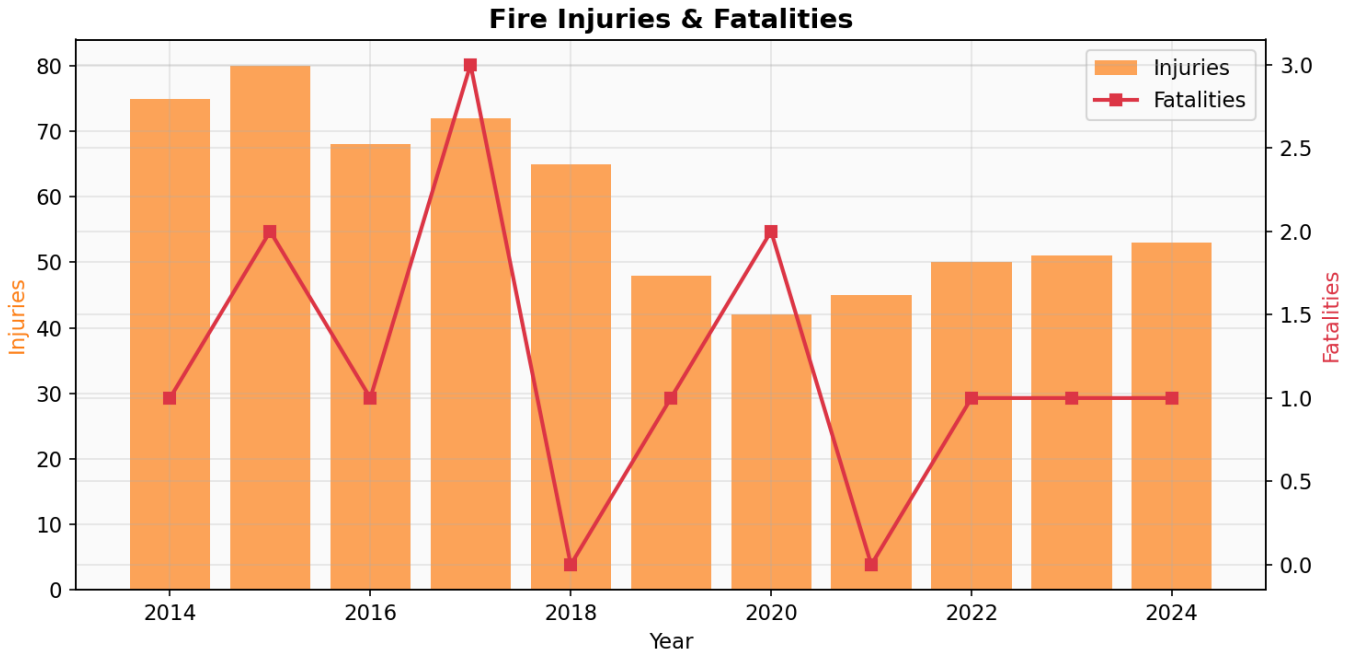
Annual fire occurrences in Singapore (Source: SCDF via Data.gov.sg)



Fire incidents by category: residential, non-residential, non-building

Fire Severity Analysis

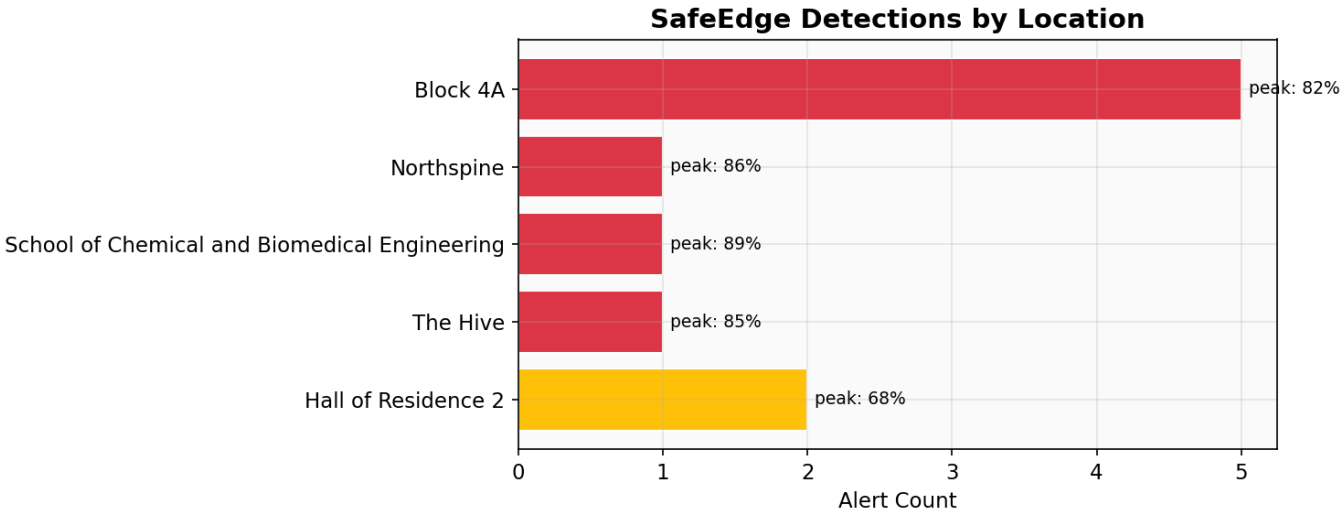
The analysis of fire severity trends reveals a concerning increase in injuries, rising from 50 in the first reporting period to 53 in the most recent, indicating a 6% escalation. Despite fatalities remaining stable at one per period, the upward trajectory of injuries underscores the critical need for improved safety measures. Implementing early detection systems, such as SafeEdge, could significantly enhance response times, potentially mitigating injury rates and preserving lives in future incidents.



Fire injuries (bars) and fatalities (line) by year

SafeEdge Detection Context

In the recent analysis of SafeEdge detections across the NTU campus, a total of 10 distinct events were recorded, with the highest concentration occurring in Block 4A and Northspine. The School of Chemical and Biomedical Engineering, The Hive, and Hall of Residence 2 also contributed to the detection count. The peak confidence level of these detections reached 88.9%, indicating a high reliability in the system's performance. Furthermore, the average detection frame rate was measured at 6.8 frames per second, suggesting a robust capability for real-time monitoring and response.



SafeEdge alerts by building location with peak confidence

Key Recommendations

In light of recent data indicating that residential fires continue to represent the highest category of incidents, particularly in kitchen areas, it is imperative to enhance fire safety measures targeting these environments. The upward trend in fire incidents post-2020 necessitates increased vigilance and proactive strategies. Implementing edge-based AI detection systems, such as SafeEdge, can significantly improve response times by identifying fires over 30 seconds earlier than traditional smoke detectors, thereby potentially saving lives and property. Furthermore, the multi-frame confirmation feature of these systems effectively minimizes false positives, while the privacy-preserving face blur technology ensures compliance with data protection regulations, making it a comprehensive solution for modern fire safety challenges.